

v1.0Muon Neutrino Deep Dive Simulation

V1.0 to V2.0 Versioning Delta — Tardigradia SubParticles Engine

Architecture: One-Muon-Neutrino Universe Architecture **Type:** Scientific Change Log | **Date:** June 2026 **Applies to:** v1.0Muon Neutrino Deep Dive Simulation V2.0.md

Revision Necessity — Four Drivers

1. **Experimental discoveries 2022-2026** — LHC Run 3, LIGO O4, KATRIN 2024, NIF ignition, and other landmark results supersede V1.0 physics parameters
2. **New confirmed particles and phenomena** — Pines’ Demon (2023), HH production evidence (2024), IceCube tau neutrino (2023), and others add entirely new simulation modes
3. **Platform upgrade** — WebGPU (Chrome 113+ 2023) replaces WebGL; enables 10^7+ particles
4. **Precision improvements** — CODATA 2022 constants, FLAG 2024 lattice QCD, NuFIT 5.3 oscillations

V1.0 vs V2.0 Specific Changes

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
Mass hierarchy	Unknown in V1.0	IceCube 2023: NH preferred at 83%; combined 3.5 σ . V2.0 defaults to Normal Hierarchy	MEDIUM	Default simulation should reflect current experimental preference
delta_CP best fit	Poorly constrained in V1.0	T2K+NOvA best fit: delta_CP ~ -135°. V2.0 PMNS matrix uses this value. 8% $\nu_\mu \rightarrow \nu_e$ asymmetry	HIGH	CP violation angle determines flavor transition rate — major physics parameter
HyperK detection events	Not covered	HyperK started 2023: HYPER-KAMIOKANDE_EVENT type when $\nu_\mu \rightarrow \nu_e$ appearance. 50 $\times$ Super-K sensitivity	LOW	Future detector mode useful for visual event display
MINERvA cross-sections	Theoretical cross-sections	MINERvA 2023: 3-5% precision NC cross-section. V2.0 interpolates from empirical table	LOW	More accurate weak interaction cross-sections

Simulation Impact Summary

The changes above drive the following modifications to the game engine architecture:

- **New particle types:** `delta_CP`
 - **Parameter updates:** All values in `static constexpr` declarations refreshed from 2022-2026 data
 - **WebGPU migration:** Particle update loop moves from CPU JavaScript to WGSL compute shaders
 - **New interaction modes:** Triggered by new experimental discoveries
-

Unchanged Architecture

All foundational philosophy is **carried forward unchanged:** - Worldline Monism (One-Particle Universe) ontology - Proper time τ as primary particle identifier - Intrinsic vs. Extrinsic property distinction - Benevolence metric ($C \cdot F \cdot S$ product formula) - Object-Oriented data structure (struct `ParticleInstance` extends to V2.0)

End of Versioning Delta | v1.0 Muon Neutrino Deep Dive Simulation V1.0 archived | V2.0 is the active specification as of June 2026